Ben Nguyen

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Education

University of Florida Gainesville, FL
B.S. Computer Engineering GPA: 3.65 *Expected May 2027*
Coursework: Data Structures & Algorithms, Microprocessor Applications, Digital Logic, Circuits I, AI for Virtual Environments

Technical Skills

Languages: C, C++, C#, Java, Python, JavaScript, AVR Assembly, VHDL, SQL
Embedded & Hardware: ESP32, Raspberry Pi Pico, Arduino, Altium Designer, I2S, Quartus Prime, SolidWorks
Software & Frameworks: Spring Boot, React, gRPC, REST APIs, WebSockets, PostgreSQL, Godot, Git
Certificates: Siemens Expedite Microcredential, AutoCAD Inventor

Projects

Portable Bluetooth Speaker – *ESP32, C++, Altium Designer, I2S, SolidWorks, PCB Design*

- Designed a custom 2-layer PCB in Altium integrating an ESP32 and dual MAX98357A class-D amplifiers, battery-powered.
- Developed C++ firmware (Arduino framework) streaming Bluetooth audio over the I2S digital protocol to dual 5W speakers, with hardware interrupts and debouncing for tactile play/pause push-buttons.
- Programmed an LCD to render real-time system status and playback metadata; modeled and 3D-printed an enclosure to house the PCB, speakers, display, and controls using SolidWorks.

Full-Stack E-Commerce Marketplace – *Java, Spring Boot, C++, gRPC, React, PostgreSQL*

- Designed and developed an e-commerce marketplace using React, Spring Boot, and PostgreSQL, featuring a custom C++ remote procedure call (gRPC) microservice for real-time dynamic pricing calculations.
- Eliminated a 96% order failure rate during high-traffic load testing by implementing optimistic locking.
- Delivered real-time order notifications via STOMP-over-WebSocket, validated the system with Mockito unit tests.

Robocode Tournament AI – *Java, Robocode API, Finite State Machines, Minimum-Risk Movement*

- Placed 1st among 26 competitors with an autonomous Java tank AI, outscoring the benchmark bot by 91%; a finite-state machine (scan/attack/evade) governed target selection, firepower, and movement policy.
- Built a minimum-risk movement engine scoring 300 sampled destinations per re-position against an inverse-square risk field weighted by relative enemy energy and firing angle, minimizing exposure to incoming fire.
- Dodged shots by detecting enemy fire from per-tick energy drops; led targets with linear-intercept aiming.

FishSchooling Simulation – *C#, Godot Engine, Git*

- Implemented flocking (alignment, cohesion, separation) on an interface-based steering system to model autonomous 2D agents; collaborated with 3 developers via Git branching and pull requests.
- Built a dynamic UI exposing fish-type-specific parameter sliders, enabling real-time behavior tuning across 5 creature types in a predator–prey ecosystem.

Experience

Machine Intelligence Laboratory (MIL), University of Florida Gainesville, FL
Electrical Team Member *August 2026 – Present*

- Developed the light board for the TerraGator rover, programming a Raspberry Pi Pico to drive AL8843SP-13 LED drivers via PWM for indicators and headlights.
- Researched PCB design practices, including component selection and layout, to inform team board designs; report progress and blockers in team standups.

UF Association of Computer Engineers Gainesville, FL
President *August 2026 – Present*

- Lead the executive board, setting semester goals and delegating events, outreach, and finances.
- Organize research lab tours and technical workshops with faculty and industry presenters.
- Serve as liaison to the ECE department and sponsors; manage the club budget.

Kyoto University — Cross-Cultural Engineering Design Program Kyoto, Japan
Exchange Student & Networking Event Coordinator *September 2025 – December 2025*

- Built FishSchooling with a mixed Japanese–international team, communicating across language barriers.
- Organized networking lunches averaging 60 attendees, raising Japanese student attendance by 40%.